FFGFT and Vopson's Computational Universe

Parallels and Differences

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https://github.com/jpascher/T0-Time-Mass-Duality

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Abstract

This document compares FFGFT (Fundamental Fractal Geometric Field Theory, T0 Time-Mass Duality) with Vopson's Computational Universe hypothesis and Second Law of Infodynamics. Both approaches start from different primitives — FFGFT from 4D torus geometry T^4, Vopson from Shannon's information theory and Landauer's principle — and arrive at structurally very similar conclusions. The agreements are numerous and substantial; the essential difference lies in the direction of derivation and in a structural disagreement about whether an outside perspective on the system is possible.

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.1 Introduction: Two Paths to the Same Destination

Melvin Vopson has been developing the *Computational Universe* hypothesis and the *Second Law of Infodynamics*: information is a physical quantity, information entropy stays the same or decreases, and gravity and symmetry can be derived from information-theoretic minimisation principles [**?**, **?**, **?**].

FFGFT begins at a different point: the conviction that no fundamental contradiction can exist between quantum mechanics and relativity, because both work too well. The search for the common geometric structure led to the 4D identification torus T^4 with the single dimensionless parameter $\xi = 4/3 \times 10^{-4}$ and the foundational relation $\tilde{T} \cdot m = 1$.

As the following analysis shows, the conclusions of both approaches are structurally identical in central respects. This is not coincidence: two independent paths arriving at the same place suggest that something is there.

.2 Parallel 1: The Minimisation Principle

Vopson's Second Law of Infodynamics

Vopson formulates: the information entropy in a closed system stays the same or decreases over time. This stands in contrast to the second law of thermodynamics (physical entropy increases). Empirical evidence: virus mutations reduce genome entropy; atoms occupy orbitals with lowest information entropy (Hund's rules) [**?**].

FFGFT: Resonant Stability as Geometric Minimisation

In FFGFT, only winding configurations with integer winding numbers $(n_\theta, n_\varphi, n_3, n_4)$ are stable — closed trajectories on T^4 resonate, open ones decay. The ξ-recursion ladder selects stable levels: only configurations satisfying the geometric resonance condition persist.

This is a geometric minimisation principle: the universe tends toward configurations that are topologically sustainable. Vopson calls this information entropy minimisation. FFGFT calls it resonant stability. The outcome is the same.

> Vopson's Second Law of Infodynamics and FFGFT's resonant stability on T^4 describe the same structural preference: the system tends toward the geometrically/information-theoretically minimal stable configurations.

.3 Parallel 2: Symmetry as Emergent Parsimony

Vopson

Symmetric structures contain less information and "save computational power". This explains why the universe prefers symmetry over asymmetry [**?**].

FFGFT

Symmetric configurations on T^4 correspond to closed winding trajectories — they minimise the geometric action and are stable fixed points of the ξ-recursion. Asymmetric configurations are either metastable or unstable.

Both conclusions: symmetry is not a coincidence but a structural preference of the underlying system.

.4 Parallel 3: Gravity as an Emergent Phenomenon

Vopson

Vopson derives Newton's law of gravity from infodynamics: objects attract each other to minimise their shared information footprint. The universe is computationally more efficient with few large objects than with billions of individual particles [**?**].

FFGFT

FFGFT derives gravity from $\tilde{T} \cdot m = 1$: the T-field couples mass to the local time structure, producing gravitational effects without a separate graviton or geometrically separately introduced curvature. The gravitational potential follows from the T-field geometry on T^4.

Two independent derivations, the same empirical law. That is the kind of convergence that is significant.

.5 Parallel 4: A Non-Directly Observable Field Explains Dark Matter Effects

Vopson

Dark matter is the stored information of the universe — the actual "program code" of reality. Not directly observable as matter, but gravitationally active and structure-bearing [**?**].

FFGFT

The T-field in FFGFT: not directly observable as matter, permeates all of space, influences gravitational dynamics via $\tilde{T} \cdot m = 1$, carries topological information through winding configurations. FFGFT does not need a separate dark matter substance because the T-field already accounts for the observed effects.

Whether one calls this "stored information" (Vopson) or "T-field from T^4 geometry" (FFGFT) — the functional description is the same.

> Vopson's "dark matter as information" and FFGFT's T-field fulfil structurally the same role: a non-directly-observable field that explains gravity and structure.

.6 Parallel 5: Information and Consciousness as Conserved Quantities

Vopson

Through the conservation laws of physics (energy and information cannot be destroyed), Vopson concludes that consciousness — or the information of a living being — persists in some form after physical death [**?**].

FFGFT

Topological invariants (winding numbers) are structurally preserved in FFGFT — even in extreme configurations such as black holes. The topological fundamental information

("Grundinformation" in the sense of Dok. 250) is not lost; it may become inaccessible to an external observer (access problem), but it continues to exist geometrically.

The conclusion is the same: information does not disappear.

.7 Parallel 6: Information Capacity of the Universe

Vopson's Bit Count

Vopson calculates the total information of the observable universe from the number of elementary particles (10^{80}) times bits per particle. Result: 10^{80} to 10^{90} bits (depending on the paper) [**?**]. These are the actually *occupied* bits.

FFGFT: Maximum Geometric Capacity

In FFGFT there is a natural upper bound via L_0:

$$L_0 = \xi \cdot \ell_P \approx 2.2 \times 10^{-39}\ \text{m} \tag{1}$$

$$N_{\text{max}}^{\text{(3D)}} = \frac{V_{\text{universe}}}{L_0^3} \approx \frac{4 \times 10^{80}\ \text{m}^3}{(2.2 \times 10^{-39})^3\ \text{m}^3} \approx 10^{195}\ \text{bits} \tag{2}$$

$$N_{\text{max}}^{\text{(4D)}} = \frac{V_{4D}}{L_0^4} \approx 10^{260}\ \text{bits} \tag{3}$$

Quantity	Value	Meaning
Vopson (occupied bits)	10^{80}–10^{90}	Actually occupied information
FFGFT (max. capacity 3D)	$\sim 10^{195}$	Geometric upper bound
FFGFT (max. capacity 4D)	$\sim 10^{260}$	T^4 capacity
Fill rate (3D)	$\sim 10^{-115}$	Universe is almost empty

No contradiction: Vopson counts the *contents*, FFGFT gives the *container size*. The universe uses a vanishingly small fraction of its geometric information capacity — consistent with the observed cosmic void.

.8 The Starting Point: Reversed Direction

The most important difference is not the result but the direction:

	Vopson	FFGFT
Primitive	Shannon information (physical)	T^4 geometry + ξ
Direction	Information → geometry emerges	Geometry → information is projection
Starting point	Landauer principle, Shannon	QM/RT unification, music theory
Gravity	From information minimisation	From $\tilde{T} \cdot m = 1$

Dark matter	Stored information	T-field from T^4

Both arrive at structurally very similar places because they describe a common deeper reality — approached from different sides.

This matches the pattern of other converging approaches in the history of science: Maxwell and Faraday arrived at electrodynamics from different sides; Heisenberg and Schrödinger developed equivalent formulations of quantum mechanics. The convergence itself is a signal.

.9 The Dissent: Is There an Outside?

Vopson's Position

If one discovered and understood the code of reality, one could theoretically "hack" it — teleportation, overcoming the speed of light, "godlike powers" [**?**]. The simulation hypothesis leaves an outside open: the universe might be running on something else.

FFGFT: Structural Exclusion

In FFGFT there is no outside to T^4. Every attempt to "hack" the code is itself a process within T^4. This is not a practical limitation — it is a geometric one. The analogy to Gödel's incompleteness theorem (Dok. 248, §9.4) is direct: a system cannot prove its own completeness from its own axioms, and there is no standpoint outside the system from which to fully describe it.

Questions like "What lies before T^4?" or "Who chose ξ?" are not well-formed in FFGFT — not because they have not yet been answered, but because they presuppose an outside perspective that structurally does not exist (cf. Dok. 248, §9).

Important: The simulation hypothesis presupposes that an "outside" exists from which the universe is simulated. FFGFT structurally excludes this: T^4 is not embedded in a larger space. "What lies outside?" is not an answerable question — analogous to "What lies north of the North Pole?"

.10 Torus Resonances as Consciousness Selection Mechanism

A further point of contact: Vopson sees the universe as an information-processing system; consciousness is part of it. In FFGFT, torus resonances are already a built-in selection mechanism:

- Only integer winding numbers are stable — resonant configurations persist, non-resonant ones decay.
- This is a geometric selection mechanism operating at all scales — from elementary particles to complex systems.
- Consciousness emerges in FFGFT terms from the recursive feedback loop (sensor → memory → movement correlation) that builds on the same topological basis.

Rudimentary forms of this self-referentiality exist already in simple T^4 configurations — analogous to Vopson's view that information is inherent in every physical structure.

Summary

Aspect	Vopson	FFGFT
Primitive	Information (physical)	T^4 geometry + ξ
Minimisation principle	2nd Law of Infodynamics	ξ-resonant stability
Symmetry	Information-minimal	Geometrically minimal
Gravity	From information minimisation	From $\tilde{T} \cdot m = 1$
Dark matter	Stored information	T-field
Information conservation	Yes	Yes (topological invariants)
Consciousness	Information preserved	Topological charge preserved
Universe bits	10^{80}–10^{90} (occupied)	10^{195} (max. capacity)
Outside perspective	Open (simulation possible)	Structurally excluded
Hackability	Theoretically possible	No — no outside

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